TECHNICAL ARCHITECTURE & COMPLETE WORKFLOW SPECIFICATION

ScrapSetu: Circular Intelligence Platform

Comprehensive Technical Approach, End-to-End Operational Lifecycle & System Architecture

📋 1. Executive Summary & Circular Economy Vision

ScrapSetu is a digital bridge and circular-economy intelligence platform designed for India's high-volume recycling ecosystem. It bridges the critical divide between domestic waste generators (households and commercial entities), informal frontline scrap collectors (*Kabadiwalas*), aggregate scrap pooling hubs, and authorized CPCB/SPCB recycling facilities.

🔍 Multimodal ScrapVision AI

Computer vision identification of scrap grade, volumetric density calculations, and transparent live Mandi rates.

🎯 Profit Optimization Matching

Algorithmic net proceed ranking factoring transport fuel, vehicle type, and authorized recycler factory rates.

📜 Cryptographic Traceability

8-step immutable digital scrap passport with weighbridge reconciliation and SHA-256 Merkle audit seals.

🔄 2. Complete End-to-End Operational Workflow (Part I)

STAGE 1 • GENERATION & INTELLIGENT BOOKING

Resident Doorstep Scrap Booking + ScrapVision AI Inspection

The resident captures a scrap photo via smartphone camera, uploads an image, or selects a certified material preset. The embedded **ScrapVision AI Neural Engine** analyzes the photo's RGB spectral distribution and geometry:

Simple Layman AI Analysis Output (English & Hindi Both):
GB *"The AI sees clean, transparent plastic bottles (PET Grade 1). Caps are separated, bottles are dry and crushed. Count: ~40 bottles. Weight: ~5.4 kg."*
IN *"एआई ने देखा: साफ और पारदर्शी प्लास्टिक की बोतलें (PET ग्रेड 1)। ढक्कन अलग हैं, बोतलें सूखी और दबी हुई हैं। लगभग 40 बोतलें। वजन: ~5.4 किलो।"*

The resident selects a preferred time slot, verifies their address, checks the live neighborhood collector radar map, and clicks **"Confirm & Broadcast Doorstep Pickup with Photo"**.

STAGE 2 • FIELD DISPATCH & VERIFICATION

Collector Job Terminal & Customer Photo Inspection

Nearby verified *Kavach* collectors receive the dispatch on their mobile terminal. Unlike traditional text-only apps, the collector inspects the **actual customer photograph**, AI volumetric weight, and distance before accepting. This eliminates wasted trips and rate disputes at the doorstep.

STAGE 3 • CORE FEATURE 1

Smart Buyer Matching (Net Profit Maximization)

Informal collectors typically sell to the nearest middleman at steep discounts. Smart Buyer Matching calculates the collector's **True Net Proceeds** by factoring road distance and transport fuel:

```
Net Proceeds (₹) = (Lot Weight × Recycler Rate) - [Base Vehicle Fare + (Distance in km × Rate/km)]
```

The system ranks authorized recyclers by net earnings, highlights the top profit-maximizing buyer, and alerts the collector if lot minimums are unmet.

🔄 2. Complete End-to-End Operational Workflow (Part II)

STAGE 4 • CORE FEATURE 2

Collaborative Scrap Pooling (Bulk Wholesale Selling)

Individual collectors rarely meet the 500 kg – 2,000 kg minimum lot sizes demanded by direct industrial recycling mills. Through Collaborative Scrap Pooling, collectors in the same cluster pool compatible scrap lots:

- **Visual Progress Bar:** Tracks pledged weight against bulk buyer capacity (e.g., 640 / 1,000 kg).
- **Wholesale Price Premium:** Direct mill sales command **+15% to +25% bonus margins** (e.g., ₹34.0/kg bulk vs. ₹28.0/kg solo rate).
- **Fairness Guarantee:** Each participant's verified weight is locked into a shared smart ledger, guaranteeing transparent proportional payout allocation upon delivery.

STAGE 5 • DOORSTEP WEIGHING & INSTANT SETTLEMENT

Calibrated TrustWeigh Hardware & Digital UPI Payment

At the doorstep, the collector pairs the certified TrustWeigh digital bluetooth scale (±0.05 kg precision). The gross weight, tare deduction, and settled Mandi rate are confirmed. The resident receives instant direct UPI payment and a digital receipt.

STAGE 6 • CORE FEATURE 3

Digital Scrap Passport & 8-Step Lifecycle Traceability

Each consigned scrap lot is issued an immutable **Digital Scrap Passport** (e.g., `WP-2024-00124`) tracking its complete lifecycle:

Lifecycle Step	Actors Involved	Verification Artefact	Compliance Status
1. Generated	Domestic Resident	Resident Profile & ScrapVision AI Log	CPCB Segregated ✓
2. Booked	Household & ScrapSetu Router	Geo-tagged Doorstep Coordinates	Logged in Dispatch Queue ✓
3. Collected	Certified Collector (Kavach)	Mobile Terminal Dispatch Ack	Vehicle Manifest Verified ✓
4. Weighed	Collector & Resident	TrustWeigh Scale Bluetooth Telemetry	Zero-Tamper Digital Scale ✓
5. Buyer Matched	Smart Matching Engine	Algorithmic Allocation Contract	Profit Optimized ✓
6. Handed Over	Collection Hub Logistics	Consignment Dispatch Gate-Pass	Transport Transit Transit ✓
7. Received	Recycler Facility Intake	50-Tonne Weighbridge Discrepancy Log	Moisture/Tare Reconciled ✓
8. Processed	Industrial Flaking/Pellet Mill	CPCB Merkle Audit Hash & SHA-256 Seal	EPR Credit Certified ✓

⚙️ 3. Short & To-The-Point Technical Approach

💻 Reactive Micro-Architecture

Built with Python and modern Streamlit, maintaining zero external paid API dependencies. Local-first session state persistence provides millisecond response times without cloud round-trip latency.

🖼️ Zero XML Parse Error Base64 Engine

All dynamic asset rendering converts SVGs and camera snapshots into RFC-2397 Base64 data URIs. Text properties are strictly escaped via `html.escape()`, preventing XML parser breakage.

🌐 Client-Side Hover Translation Injector

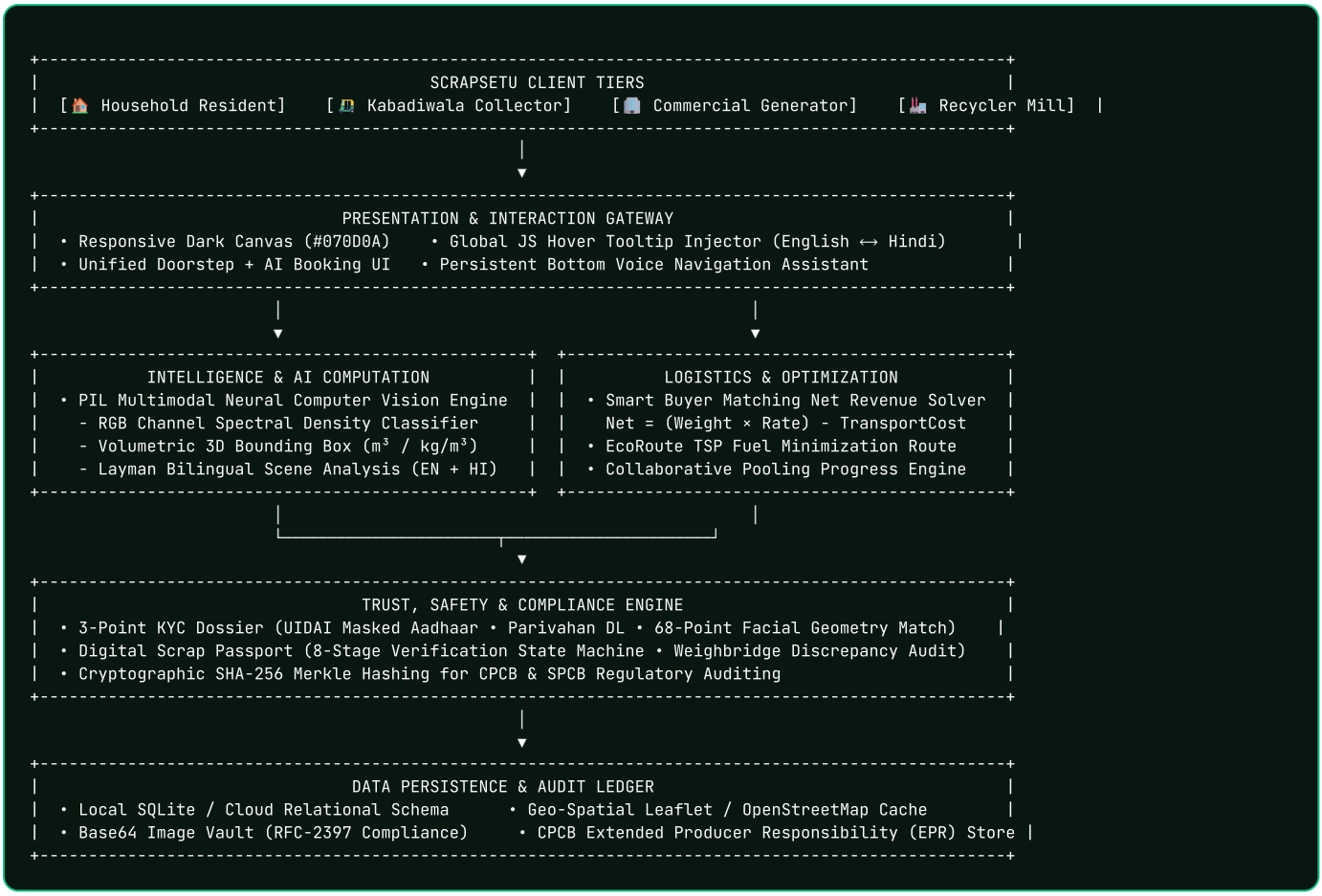
Strict single-language primary rendering eliminates visual bilingual clutter. A continuous DOM observer binds hover tooltips to tabs and buttons, showing instant alternate-language translations.

🧭 Pinned Bottom Navigation Assistant

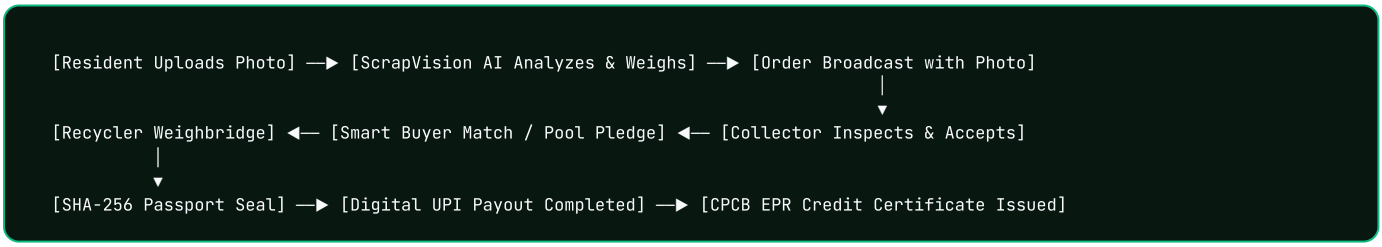
Pinned directly at the page footer, providing voice phrase simulation and deterministic redirection across all 5 user roles and 13 platform sub-modules.

📐 4. Complete System Architecture & Data Flow

The diagram below illustrates the 5-tier architecture connecting the presentation layer, intelligence micro-services, and immutable audit logs:



Entity State Transition & Workflow Lifecycle



📊 5. Summary Matrix: Technical Innovation vs. Traditional Scrap Market

Functional Pillar	Traditional Scrap Trade	ScrapSetu Circular Platform
Material Inspection	Manual guesswork; frequent doorstep disputes.	**ScrapVision AI:** Instant grade detection & volumetric weight calculation.
Collector Dispatch	Random shouting on streets; blind bookings.	**Photo-Attached Terminal:** Full inspection before order acceptance.
Selling Margins	Distress selling to local intermediary cartels.	**Smart Buyer Matching:** Direct recycler matching maximizing net proceeds.
Bulk Sales Access	Small collectors locked out of industrial mills.	**Collaborative Scrap Pooling:** Combined lots earning +15% to +25% bonus.
Regulatory Proof	Paper receipts; zero verifiable environmental proof.	**Scrap Passport:** Cryptographic SHA-256 Merkle trail with CPCB compliance.